深度學習
Deep Learning

Vision Transformer (ViT)、自監督式學習與預訓練模型

Instructor: 林英嘉 (Ying-Jia Lin)

2025/11/12

[Course GitHub]

[Slido # DL1112]

Announcements

Project checkpoints

- ~~Week 9: 確定各組的題目~~
- Week 12: 進度報告 PPT (5 pages)
- Week 14: 進度報告 PPT (5+5 pages), Presentations (selected teams)
- Week 16: Final presentations for all teams
- Week 16-17: 繳交書面報告以及程式碼

Project 各個階段分數佔比

Final Project 佔學期總成績 40%

查核點 (週次)	對象: 繳交內容	分數佔比
Checkpoint1 (Week 12)	All teams: 進度報告 PPT (5 pages)檔案	5%
Checkpoint2 (Week 14)	All teams: 進度報告 PPT (5+5 pages*)檔案 Selected teams: 取6組 (1題目2組) 於課堂中報告，1組 10min	5%
Checkpoint3 (Week 16)	All teams: 最終口頭報告	15%
Checkpoint4 (Week 16-17)	All teams: 書面報告檔案 (含程式碼)、小平台 (optional)	15%

*繼承Checkpoint1內容+實作

Week 12 要繳交什麼？

- 清晰易懂的簡報 5 頁，一組繳交一份
  - 介紹競賽任務
  - 舉例說明資料的長相
  - 做簡單的資料分析
    - 可以參考 Kaggle Code，關鍵字: EDA (Exploratory Data Analysis)
      - 請附上來源!!
  - 預期的實作方法
    - 預期實作/使用的模型
      - 可以參考 Kaggle Code，請附上來源!!
  - (預期) 分工模式

Week 12 繳交注意事項

- 5 頁是最低需求 (不包含首頁跟最後一頁)
  - 可以多於5頁!!
- 一組繳交一份，請上傳至 E-Learning
- Deadline: 2025/11/21 23:59
- 檔名：DL_teamN_checkpoint1.pdf

Outline

- Vision Transformer (ViT)
- **Self-supervised learning**
  - Images
    - Autoencoders
    - Contrastive Learning
  - Text
    - Next-token prediction (訓練方式為 Language Modeling)
    - Cloze test (訓練方式為 Masked-language Modeling)

Transformer能不能用在影像上呢？

Vision Transformer (ViT)

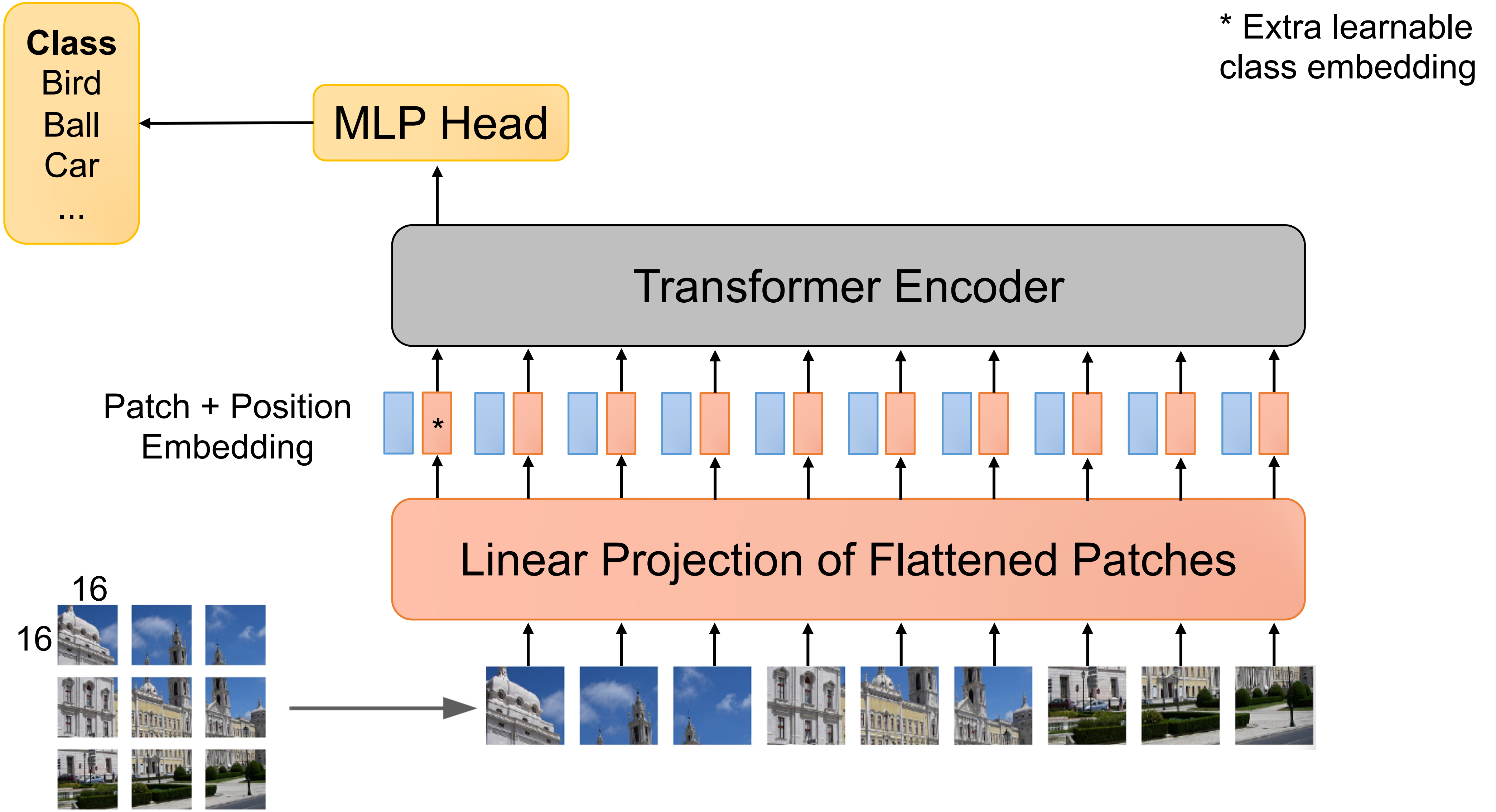
Concept: 把影像區域當成 token 輸入

Published as a conference paper at ICLR 2021

AN IMAGE IS WORTH 16X16 WORDS: TRANSFORMERS FOR IMAGE RECOGNITION AT SCALE

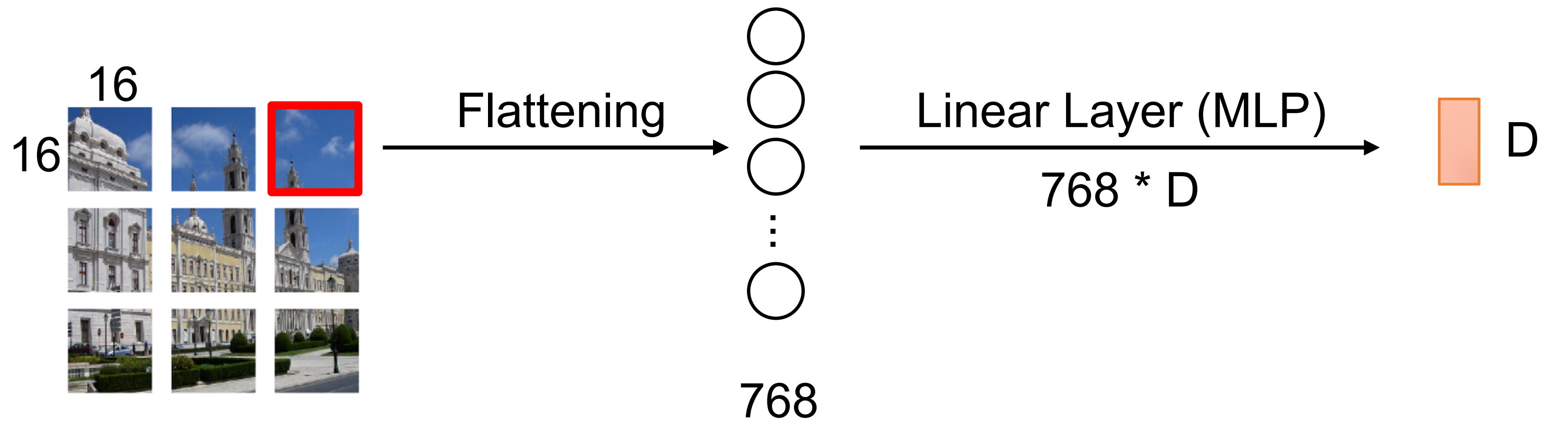
Alexey Dosovitskiy[*,†], **Lucas Beyer**[*], **Alexander Kolesnikov**[*], **Dirk Weissenborn**[*], **Xiaohua Zhai**[*], **Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, Neil Houlsby**[*,†]
[*]equal technical contribution, [†]equal advising
Google Research, Brain Team
{adosovitskiy, neilhoulsby}@google.com

Vision Transformer (ViT)



Linear Projection of Flattened Patches

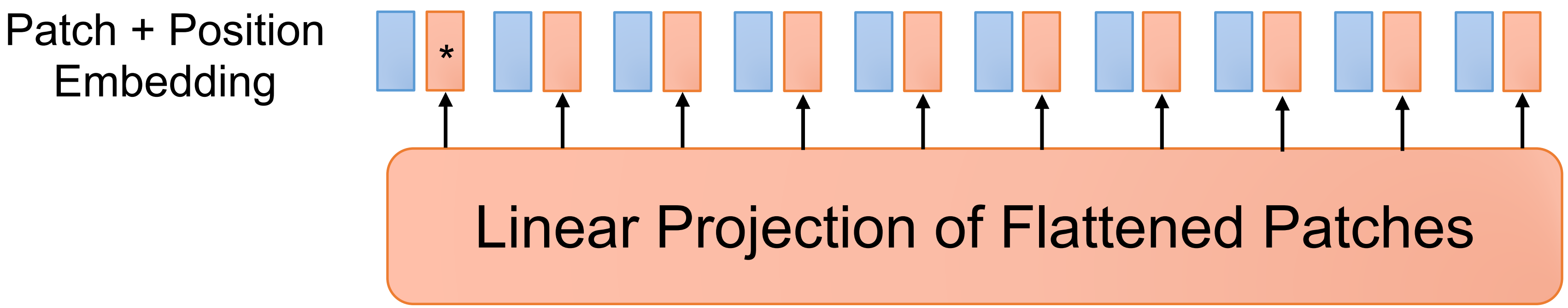
- 假設原始輸入是 224*224 的影像，切成 16*16 的小張影像 (patch)
- 每張 patch 進行 flattening 後，數值總數：16 * 16 * num_channels
- 使用線性轉換層將影像原始資料處理為 image embeddings
  - 輸出的維度為 D，Linear Projection 的矩陣大小為 16 * 16 * num_channels * D

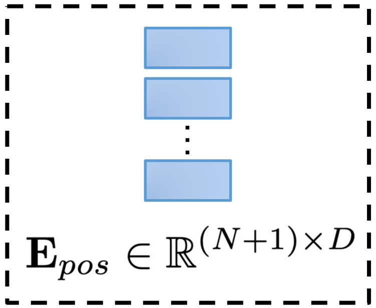


Positional Embedding in ViT

* Extra learnable class embedding (CLS token)

- 假設原始輸入是 224*224 的影像，切成 16*16 的 patches
  - num_patches = 14 * 14 = 196，等同於有 196+1 (CLS) 個 embeddings
  - 每個 patch 各自的 embedding (維度為D) 會在訓練過程中自動更新
    - 絕對位置資訊編碼 (Absolute positional Encoding)

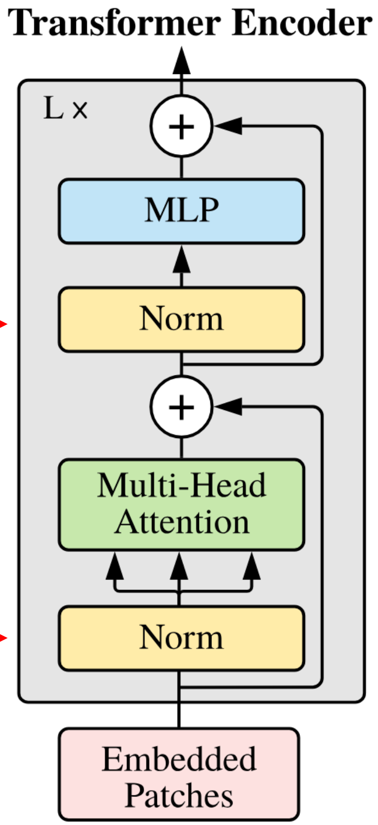




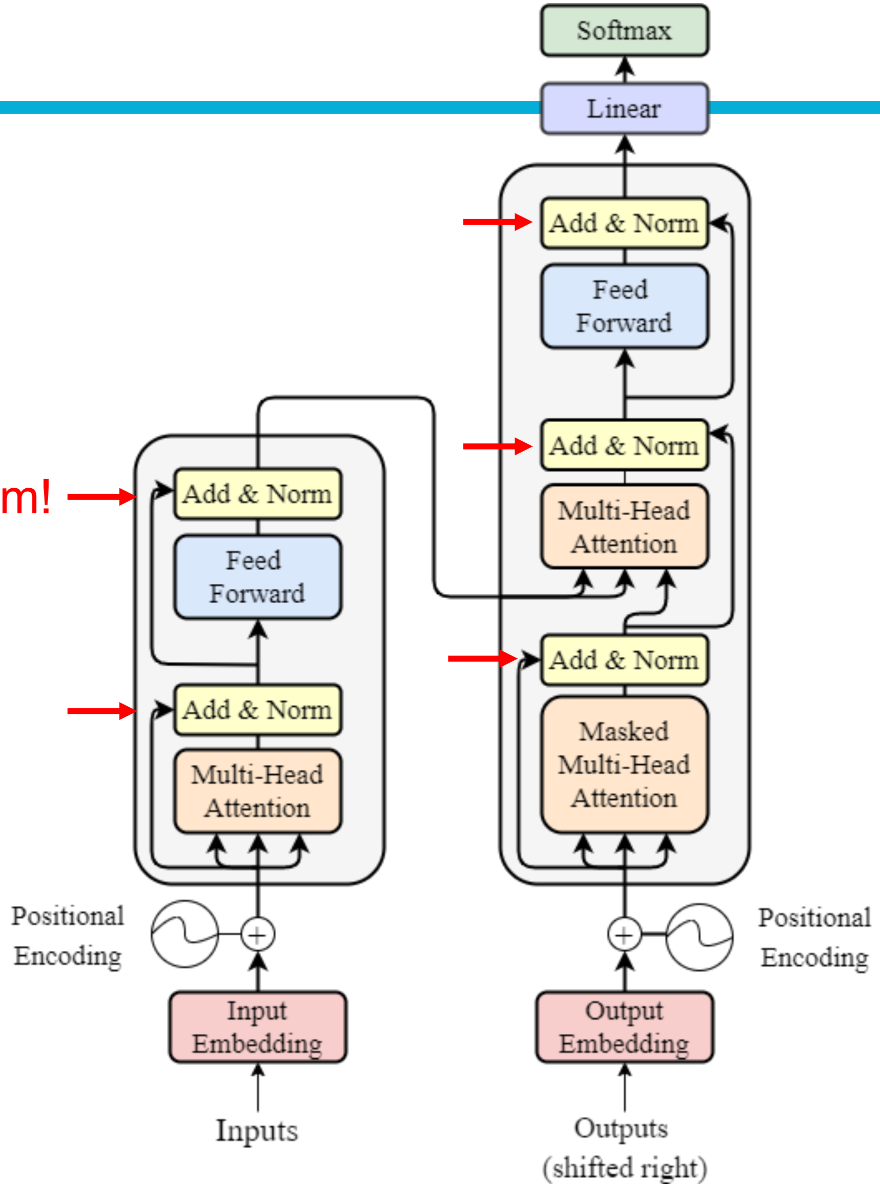
ViT 使用 Pre-norm

Left: ViT Transformer layer
Right: 原始 Transformers

使用 Pre-norm!

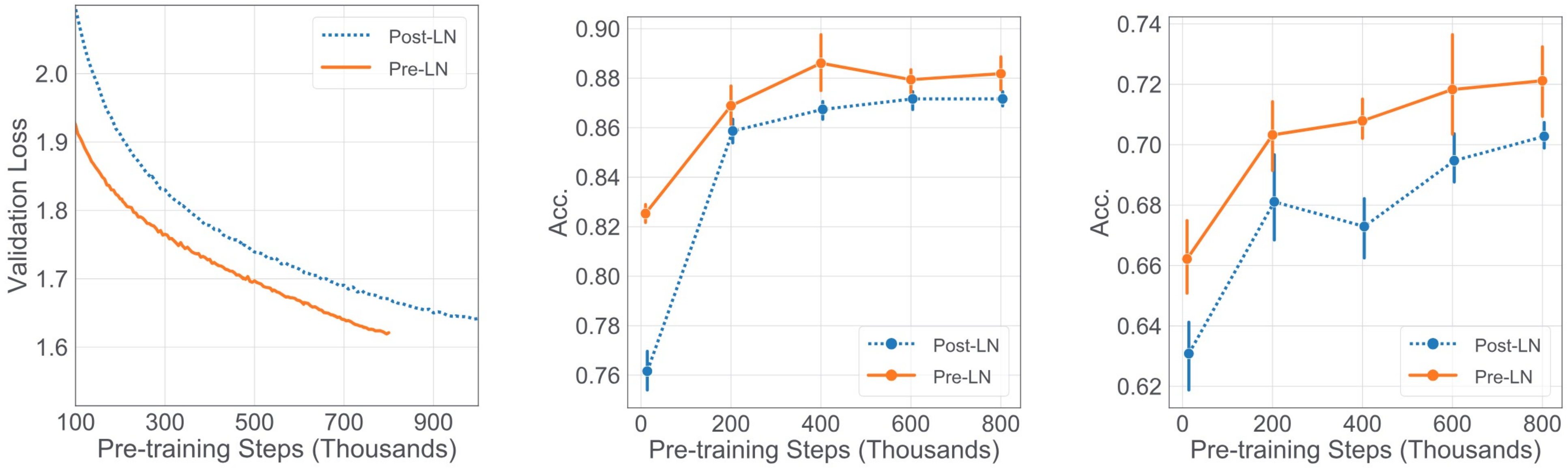


使用 Post-norm!



Why using Pre-Norm (Pre-Normalization)?

Xiong, Ruibin, et al. "On layer normalization in the transformer architecture." International conference on machine learning. 2020.



(a) Validation Loss on BERT (b) Accuracy on MRPC (c) Accuracy on RTE

Figure 5. Performances of the models on unsupervised pre-training (BERT) and downstream tasks

Self-supervised Learning

簡稱SSL

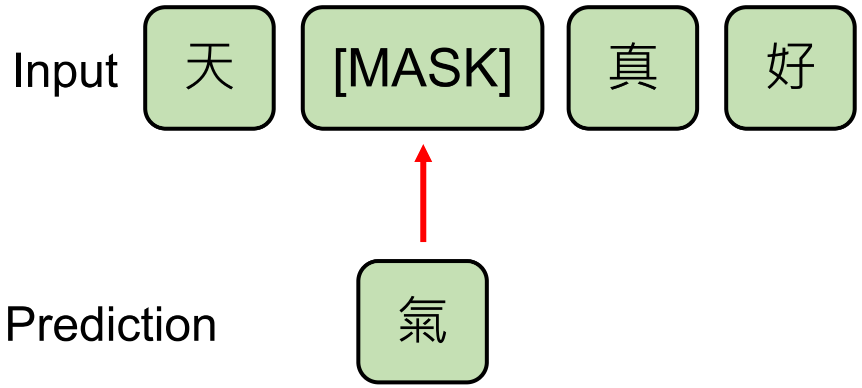
Why self-supervised?

- 深度學習模型需要大量資料+答案才能進行訓練

哪裡來？標資料很貴，最好資料本身就有答案

影像處理SSL例子

自然語言處理SSL例子



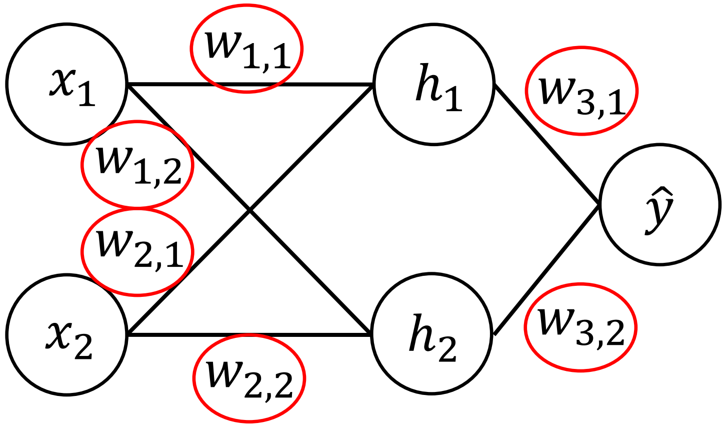
先 Pre-training，再 Fine-tuning

Pre-training → **Fine-tuning**

在大量資料上進行訓練，通常是自監督式 (Self-Supervised Training)

在**目標**資料上 (Down-stream tasks, 下游任務) 進行訓練，通常是監督式 (Supervised Training)，也就是需要有標註的資料才能進行模型訓練

Pre-training 和模型初始化有關



參數可能為隨機初始化

$$h_1 = w_{1,1}x_1 + w_{2,1}x_2 + b_1$$

$$h_2 = w_{1,2}x_1 + w_{2,2}x_2 + b_2$$

$$\hat{y} = w_{3,1}h_1 + w_{3,2}h_2 + b_3$$

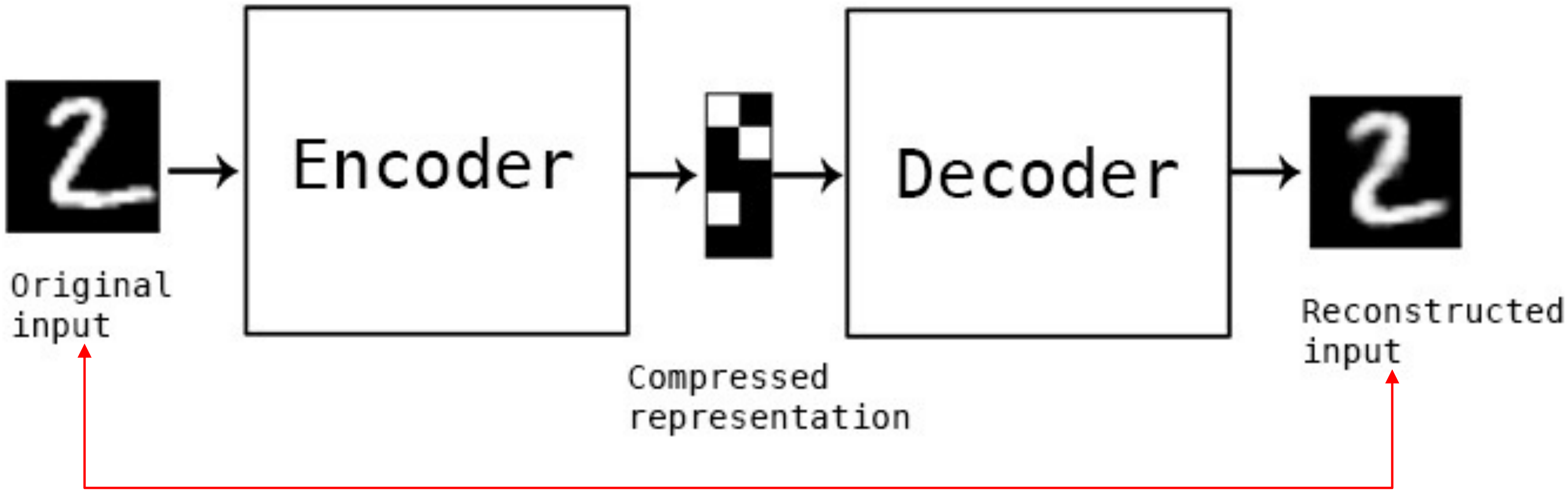
💡預訓練就是學習一個有意義的初始化點，而非隨機點。

影像處理 SSL

Autoencoder (自動編碼器, AE)

以最原始的 Autoencoder 為例

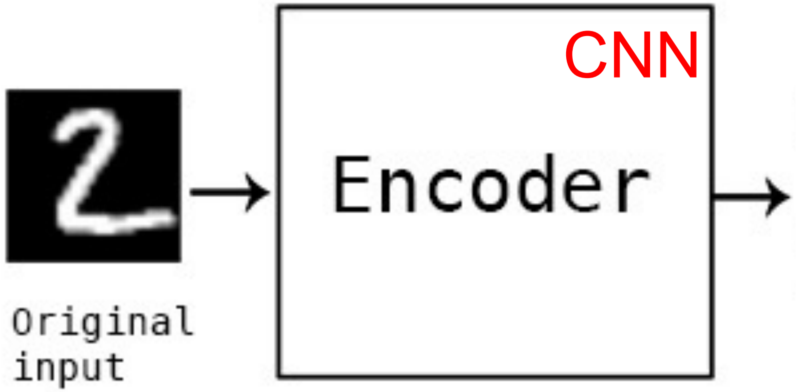
Autoencoder 的訓練方式即為 self-supervised learning



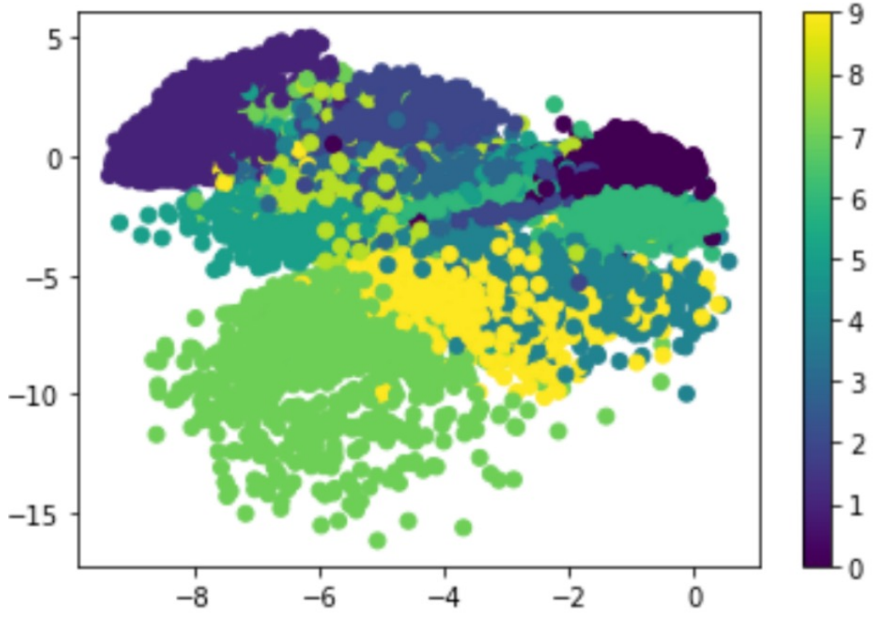
訓練目標：Reconstruction error (MSE, Mean Squared Error) 越低越好

Figure source: https://blog.keras.io/img/ae/autoencoder_schema.jpg

AE 的功能1：降維



[2, 1 ,1]
(C, H, W)

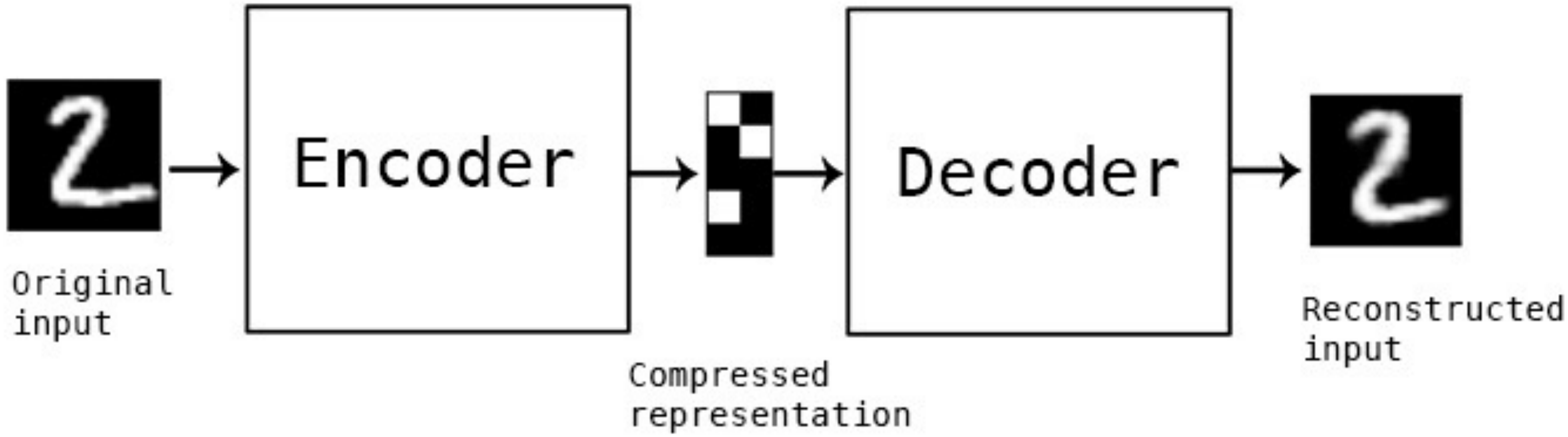


AE 與 PCA (主成分分析)
t-SNE (t-distributed Stochastic Neighbor Embedding)
同為降維 (dimensionality reduction) 作法

Figure source:
Left: https://blog.keras.io/img/ae/autoencoder_schema.jpg
Right: https://jason-chen-1992.weebly.com/home/-autoencoder

AE 的功能2：得到一個好的初始 Encoder

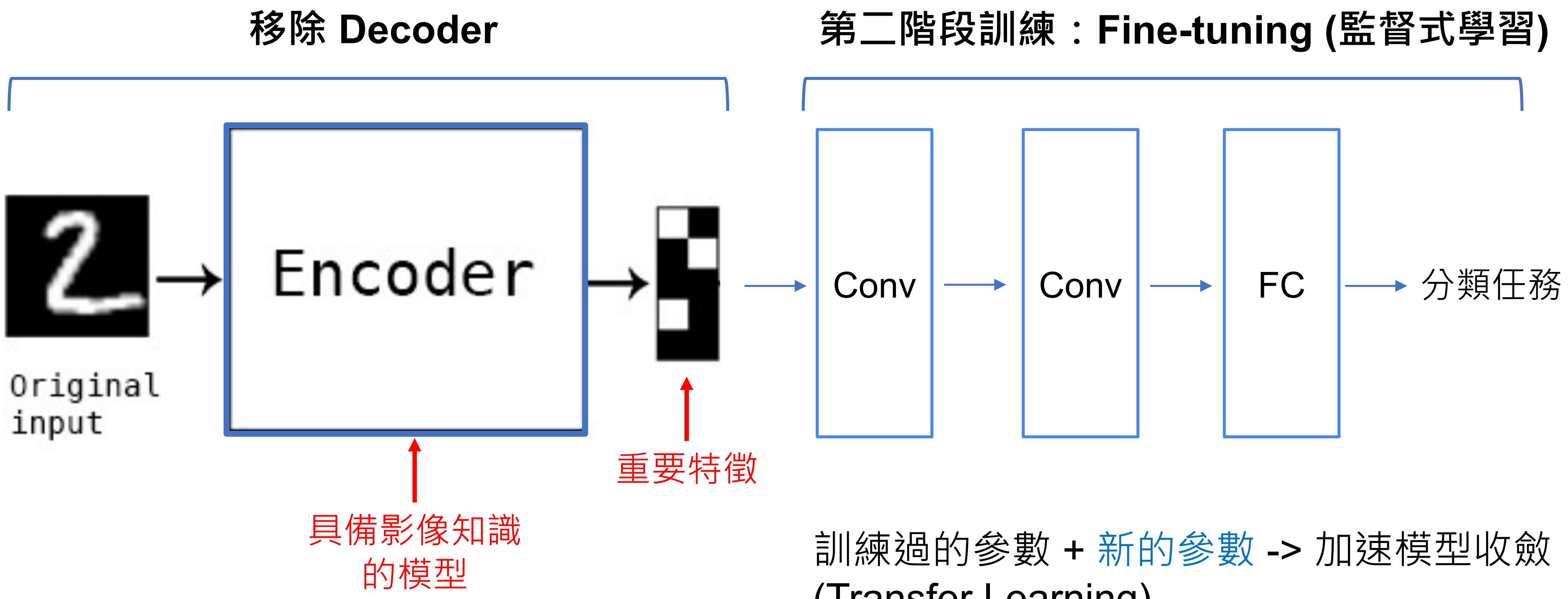
第一階段訓練：**Pre-training (SSL)**



AE 的功能2：得到一個好的初始 Encoder

將 AE 的 Encoder 當作特徵萃取器，遷移到下游分類任務

Conv: convolutional layer
FC: fully-connected layer



訓練過的參數 + 新的參數 -> 加速模型收斂
(Transfer Learning)

https://arxiv.org/abs/2201.03898

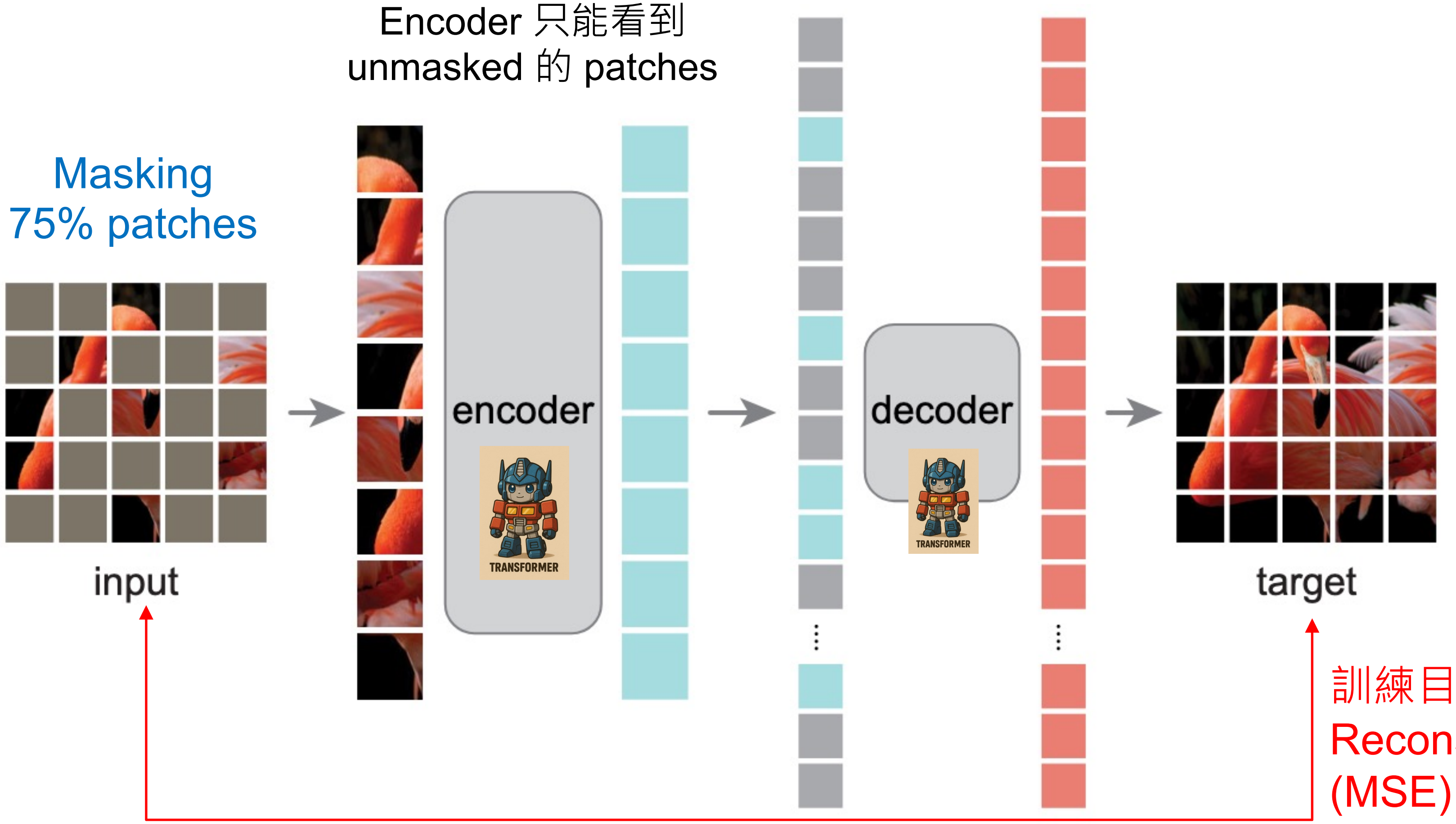
Masked autoencoders (MAE)

Masked patches 加回去 encoder output 後，作為 decoder input

Encoder 只能看到 unmasked 的 patches

Masking 75% patches

把圖片切成 patches (例如 ViT：16x16)



訓練目標：Reconstruction error (MSE) 越低越好

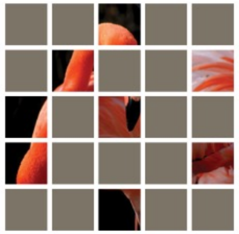
He, Kaiming, et al. "Masked autoencoders are scalable vision learners." CVPR 2022.

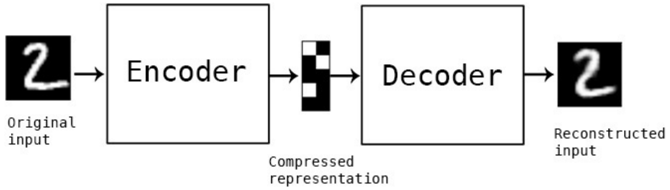
MAE 細節

- Decoder 知道 patches 的順序 (位置資訊) 嗎？
  - Yes，且 Encoder 也有位置資訊 (同 ViT)
- Decoder 的 input 包含了 Encoder outputs 和 masked patches，那這些會照順序排好再輸入給Decoder嗎？
  - Yes

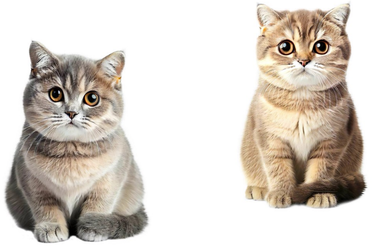
不同的 SSL 策略

學習尋找影像內部
的關聯性





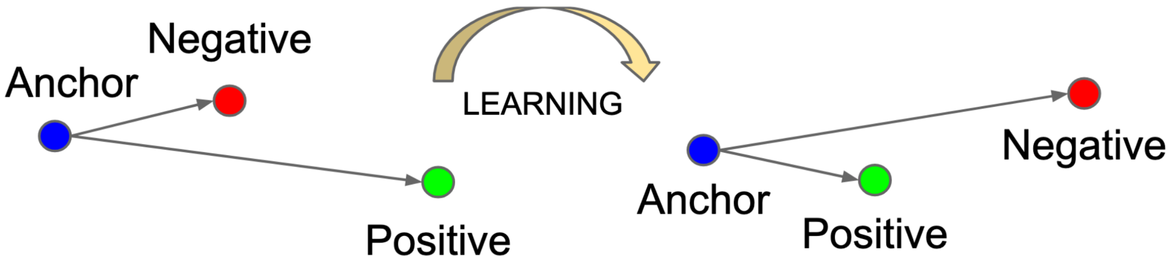
學習尋找影像之間
的關聯性

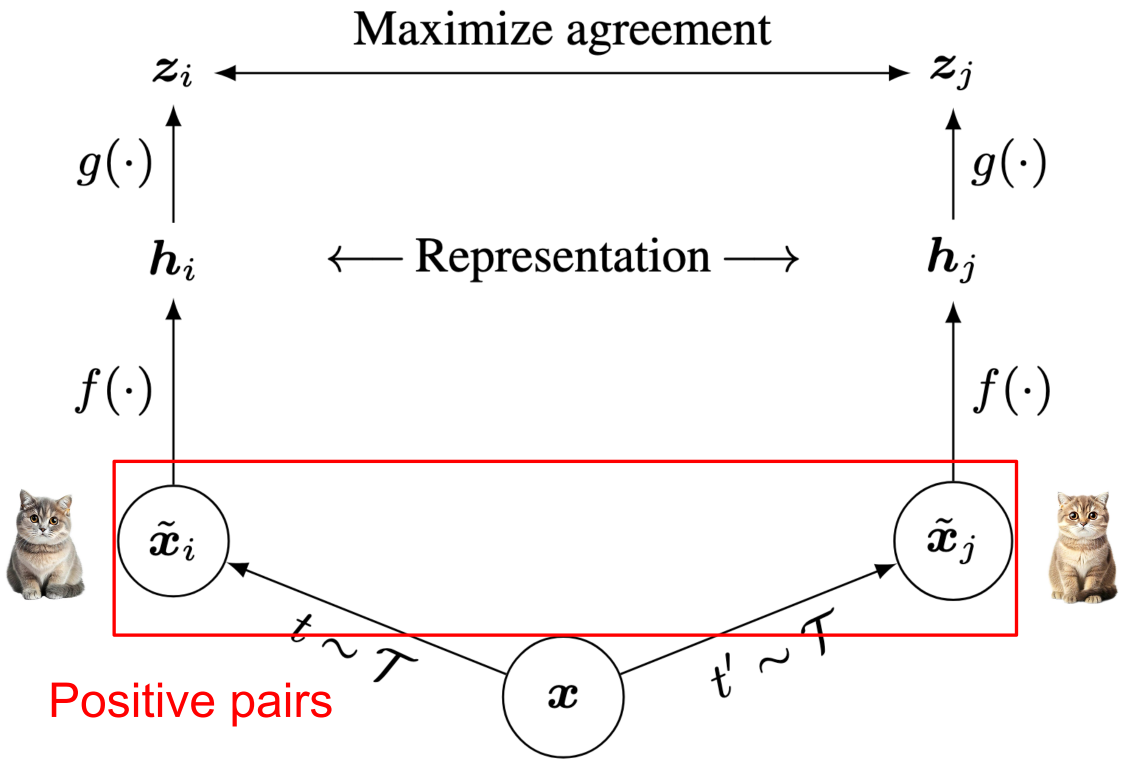


Contrastive Learning in Images

f (·): encoder network
g(·): projection head
z: output logits

- The concept of contrastive learning is to maximize the agreement of positive pairs [1][2] .
- A model will be able to learn the connections among similar data and dissimilar data.





[1] Left Figure source: Schroff, Florian, Dmitry Kalenichenko, and James Philbin. "Facenet: A unified embedding for face recognition and clustering." CVPR 2015.

[2] Right Figure source: Chen, Ting, et al. "A simple framework for contrastive learning of visual representations." ICLR 2020.

Image data augmentation approaches

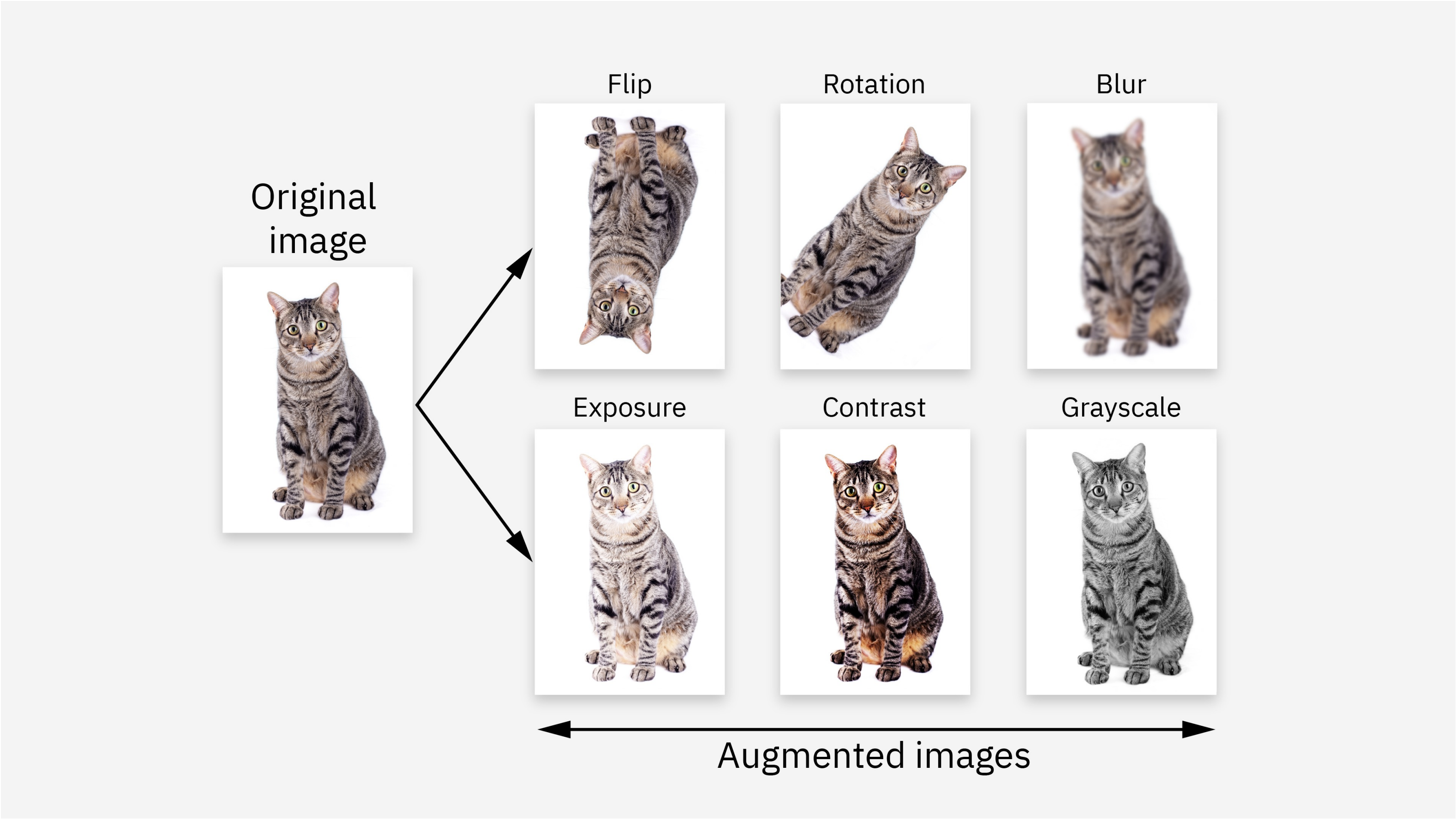
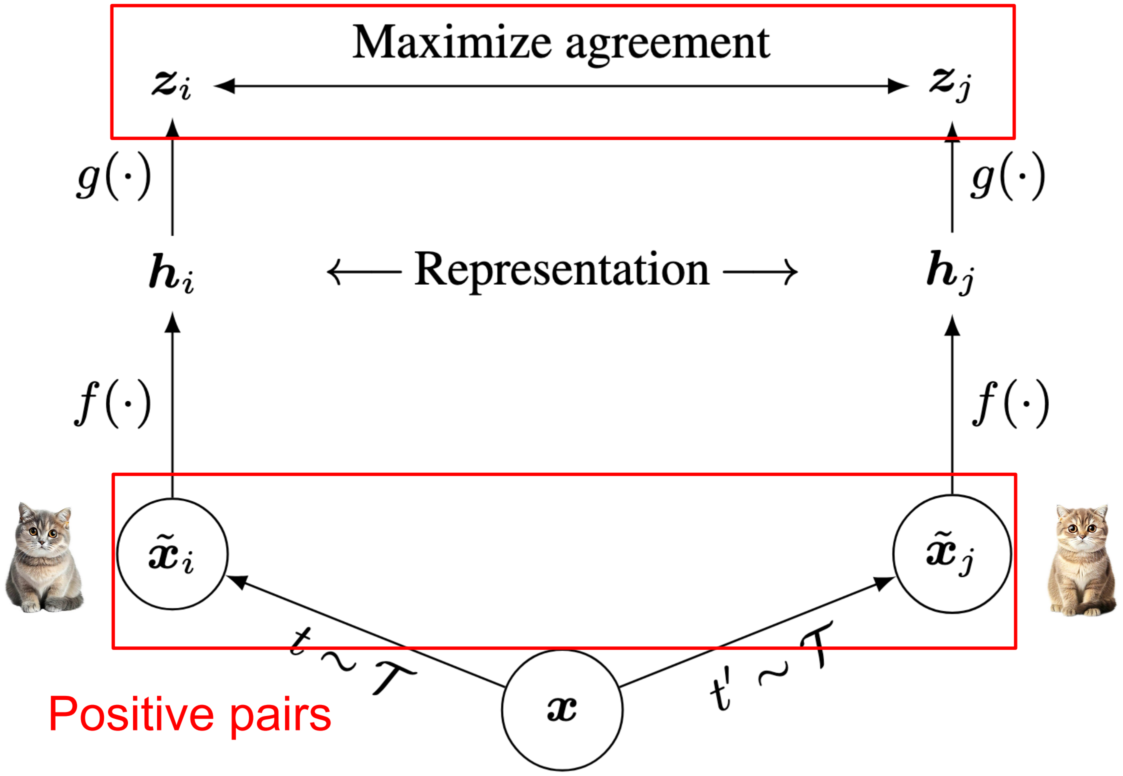
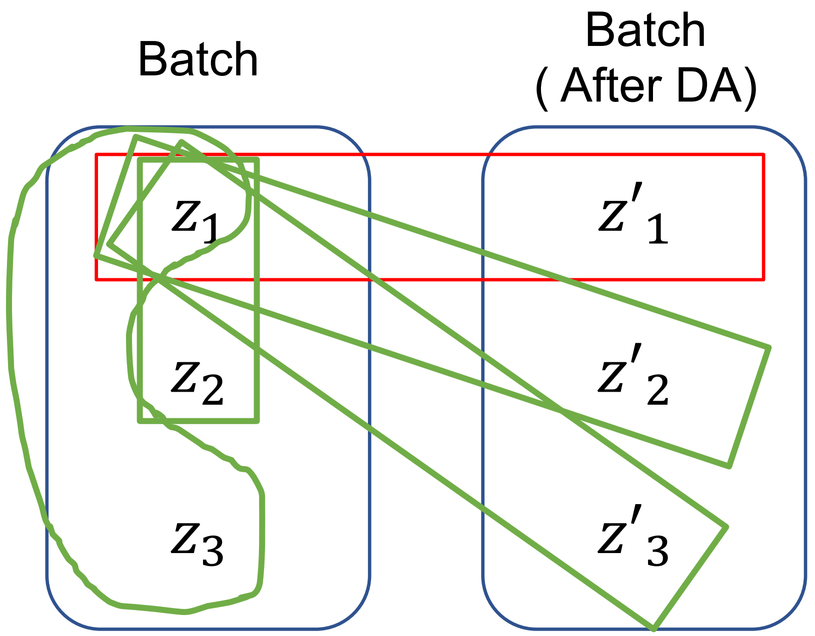


Figure source: https://www.ibm.com/think/topics/data-augmentation

Contrastive Learning Loss (1)

DA: data augmentation
假設 Batch_size 為 N



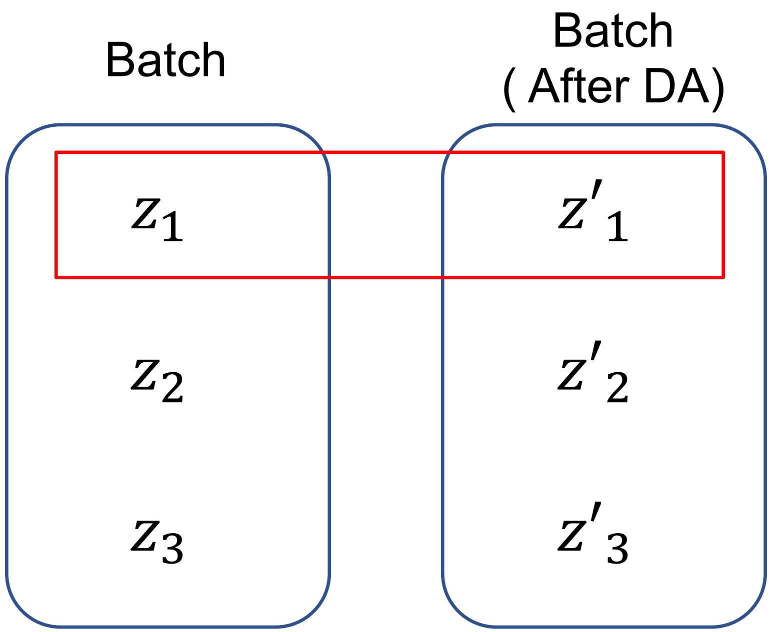


以x_1為例：
正樣本對 -> (z_1, z'_1)
負樣本對 -> $\boxed{(z_1, z'_2)、(z_1, z'_3)、(z_1, z_2)、(z_1, z_3)}$

Chen, Ting, et al. “A simple framework for contrastive learning of visual representations (SimCLR)." ICLR 2020.

Contrastive Learning Loss (2)

z 為模型輸出的 representation
假設 Batch_size 為 N



	取 Softmax 後之機率	正確答案
$\mathrm{sim}(z_1, z'_1)$	0.3	1
$\mathrm{sim}(z_1, z'_2)$	0.2	0
$\mathrm{sim}(z_1, z'_3)$	0.1	0
$\mathrm{sim}(z_1, z_2)$	0.1	0
$\mathrm{sim}(z_1, z_3)$	0.3	0

→ Cross-entropy

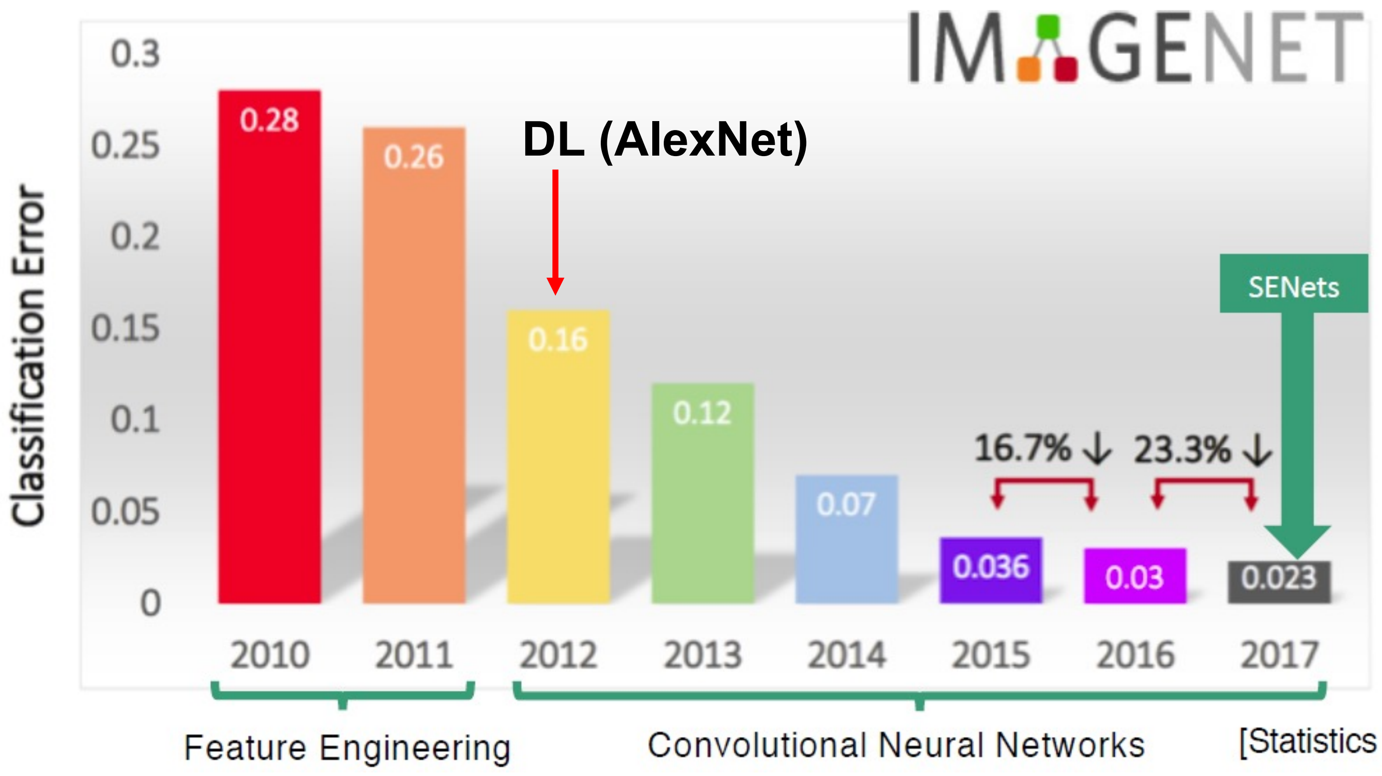
以x_1為例：
正樣本對 -> (z_1, z'_1)
負樣本對 -> (z_1, z'_2)、 (z_1, z'_3)、(z_1, z_2)、 (z_1, z_3)

相當於在做 2N-1 個類別之分類任務

Chen, Ting, et al. "A simple framework for contrastive learning of visual representations (SimCLR)." ICLR 2020.

ImageNet Competition

Complete name: ImageNet Large Scale Visual Recognition Challenge (ILSVRC)



[Statistics provided by ILSVRC]

Figure source: https://www.kaggle.com/discussions/getting-started/149448

ImageNet (有效，但非SSL)

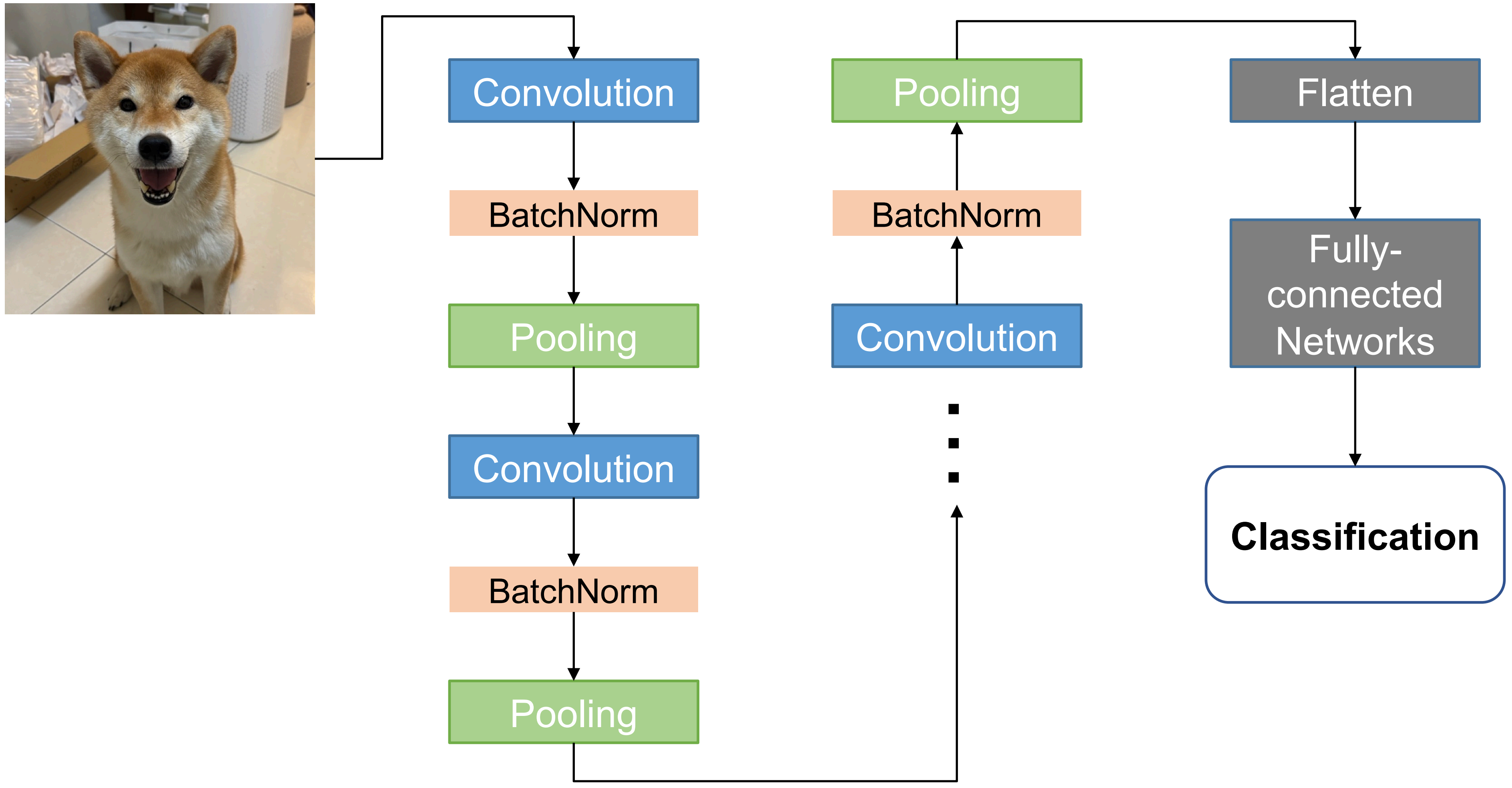
- 剛剛講了很多基於 SSL 的 Representation Learning
  - 含 Autoencoders, contrastive learning
- ImageNet pre-training 是深度學習中最早、最有效的 representation learning 策略之一
  - 直接標大量資料的答案 (約120萬張)

ImageNet pre-trained models on PyTorch

Weight	Acc@1	Acc@5	Params	GFLOPS	Recipe
AlexNet_Weights.IMAGENET1K_V1	56.522	79.066	61.1M	0.71	link
ConvNeXt_Base_Weights.IMAGENET1K_V1	84.062	96.87	88.6M	15.36	link
ConvNeXt_Large_Weights.IMAGENET1K_V1	84.414	96.976	197.8M	34.36	link
ConvNeXt_Small_Weights.IMAGENET1K_V1	83.616	96.65	50.2M	8.68	link
ConvNeXt_Tiny_Weights.IMAGENET1K_V1	82.52	96.146	28.6M	4.46	link
DenseNet121_Weights.IMAGENET1K_V1	74.434	91.972	8.0M	2.83	link
DenseNet161_Weights.IMAGENET1K_V1	77.138	93.56	28.7M	7.73	link
DenseNet169_Weights.IMAGENET1K_V1	75.6	92.806	14.1M	3.36	link
DenseNet201_Weights.IMAGENET1K_V1	76.896	93.37	20.0M	4.29	link
EfficientNet_B0_Weights.IMAGENET1K_V1	77.692	93.532	5.3M	0.39	link
EfficientNet_B1_Weights.IMAGENET1K_V1	78.642	94.186	7.8M	0.69	link
EfficientNet_B1_Weights.IMAGENET1K_V2	79.838	94.934	7.8M	0.69	link
EfficientNet_B2_Weights.IMAGENET1K_V1	80.608	95.31	9.1M	1.09	link
EfficientNet_B3_Weights.IMAGENET1K_V1	82.008	96.054	12.2M	1.83	link
EfficientNet_B4_Weights.IMAGENET1K_V1	83.384	96.594	19.3M	4.39	link
EfficientNet_B5_Weights.IMAGENET1K_V1	83.444	96.628	30.4M	10.27	link
EfficientNet_B6_Weights.IMAGENET1K_V1	84.008	96.916	43.0M	19.07	link
EfficientNet_B7_Weights.IMAGENET1K_V1	84.122	96.908	66.3M	37.75	link
EfficientNet_V2_L_Weights.IMAGENET1K_V1	85.808	97.788	118.5M	56.08	link
EfficientNet_V2_M_Weights.IMAGENET1K_V1	85.112	97.156	54.1M	24.58	link
EfficientNet_V2_S_Weights.IMAGENET1K_V1	84.228	96.878	21.5M	8.37	link
GoogLeNet_Weights.IMAGENET1K_V1	69.778	89.53	6.6M	1.5	link

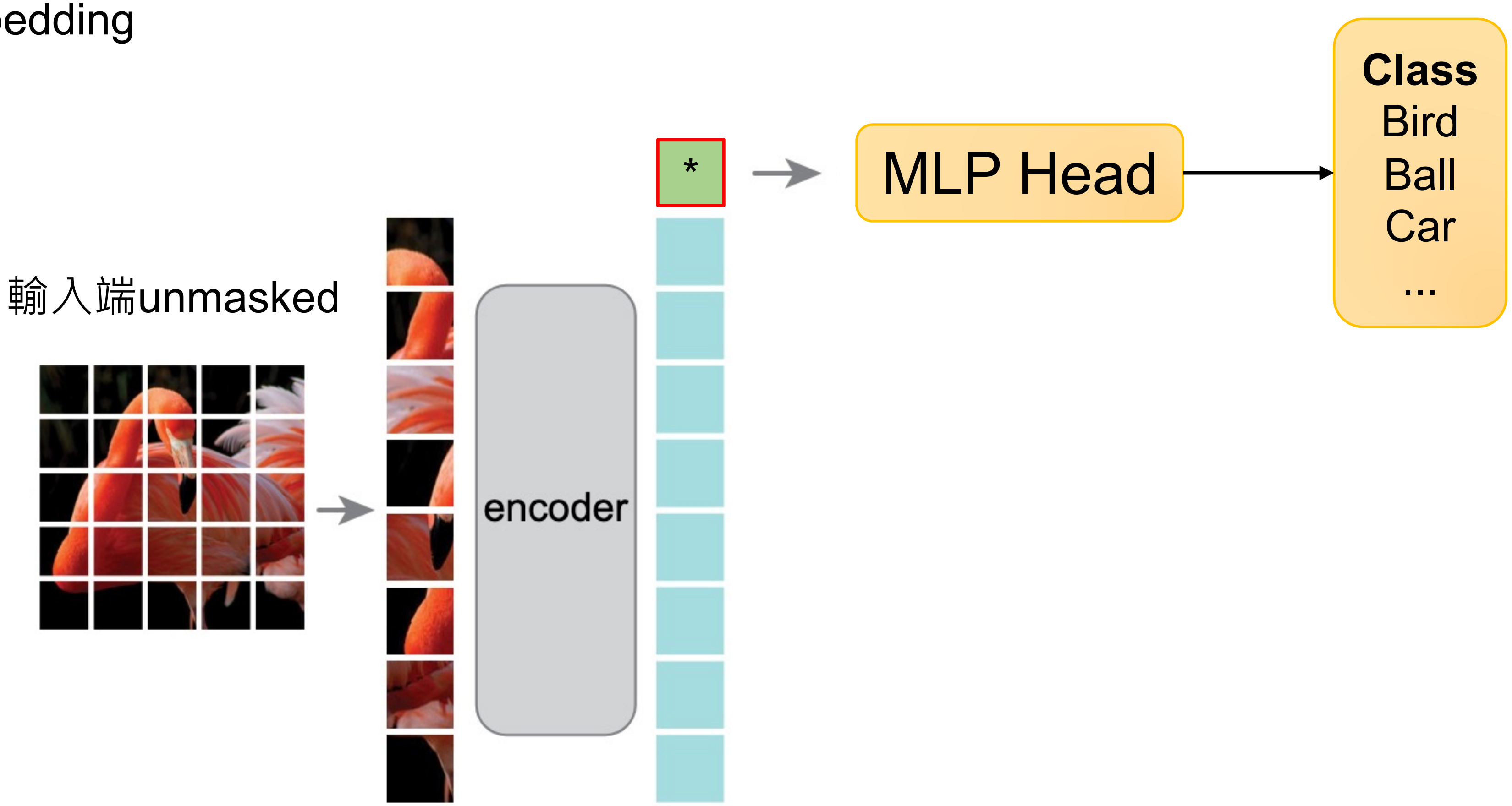
https://pytorch.org/vision/main/models.html

[Recap] (Supervised) Fine-tuning with CNN



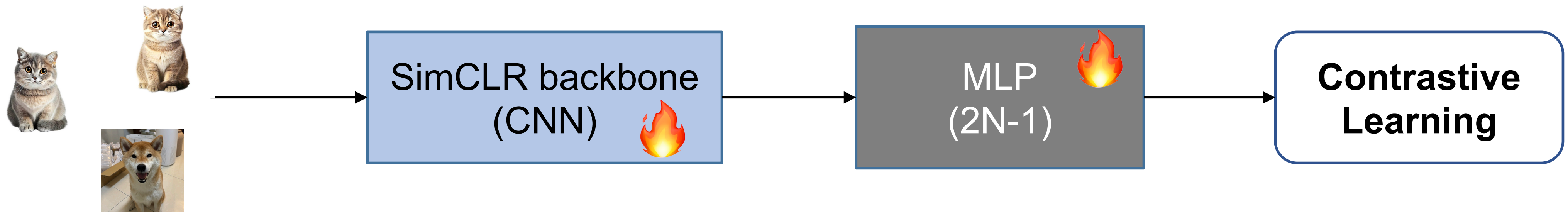
(Supervised) Fine-tuning with MAE

* Extra learnable
class embedding

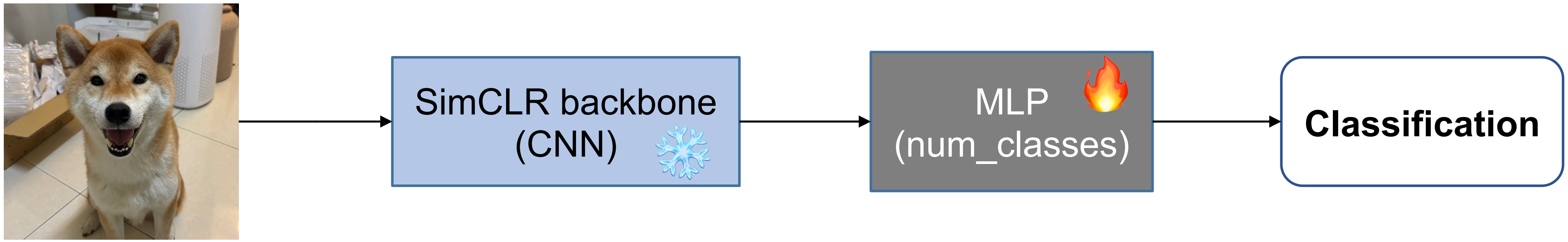


(Supervised) Fine-tuning with SimCLR

Pre-training



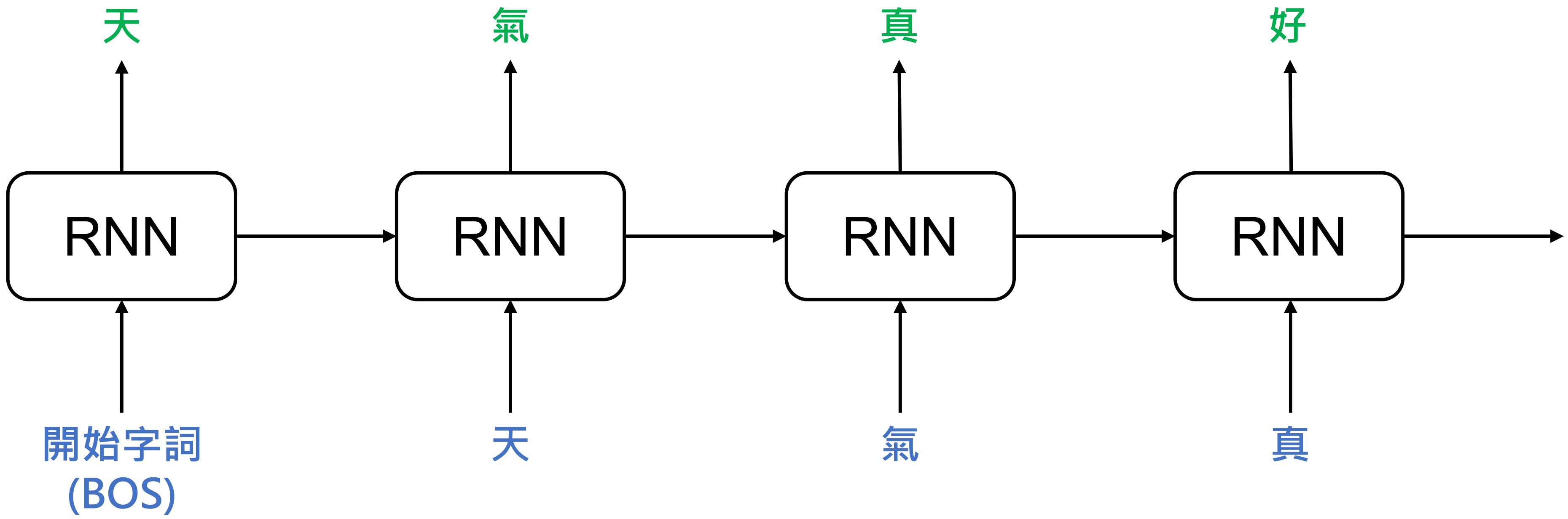
Fine-tuning



自然語言處理 SSL

條件機率

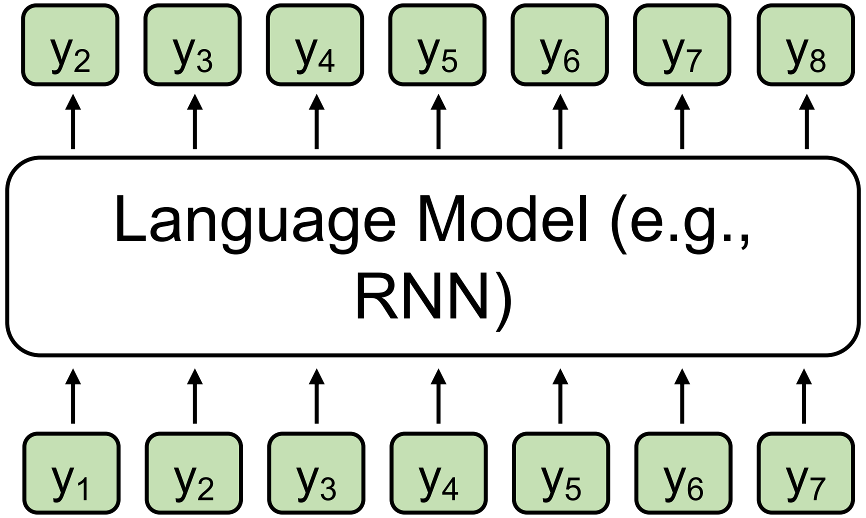
假設要讓模型生成一個句子：「天氣真好」



P(天|BOS)　P(氣|開始字詞、天)　P(真|開始字詞、天、氣)　P(好|開始字詞、天、氣、真)

Next-token prediction

Next-token prediction: 在給定上下文序列 $y_1, y_2, \dots, y_{t-1}$，模型在第 t 個時間點會預測下一個字生成的機率值

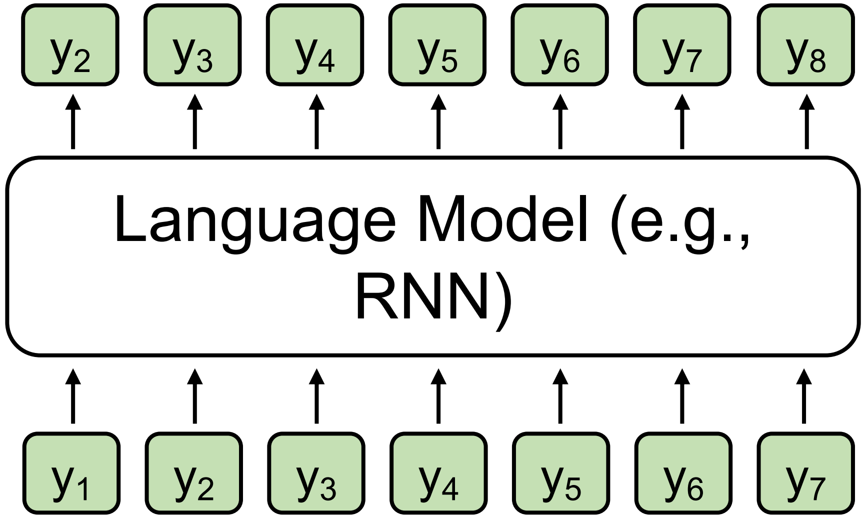


$$P(y_t | y_1, y_2, \dots, y_{t-1})$$

- The task involving predictions of upcoming words is next-token prediction.

模型如何進行訓練？

Language Modeling: 綜和每個時間點預測的聯合機率分佈 (每個時間點的機率值相乘)



$$\prod_{t=1}^{n} P(y_t|y_1, y_2, \dots, y_{t-1})$$

- 語言模型：能預測語言序列生成之機率分布的模型

Language Modeling and Cross-entropy (1)

為了使語言模型能夠以分類的形式被訓練，**通常會取log**

$$\log\left(\prod_{t=1}^{n} P(y_t | y_1, y_2, \dots, y_{t-1})\right)$$

$$= \sum_{t=1}^{n} \log P(y_t | y_1, y_2, \dots, y_{t-1})$$

加上負號之後就是 **Cross-entropy**

Language Modeling and Cross-entropy (2)

Language modeling 在 t 時間點的正確答案是 t+1 的字詞

I love CGU

t t+1

Example Vocab (假設只有5個字)

Token	Logits	取 Softmax 後之機率	正確答案
CGU	0.0011	0.78	1
I	0.0012	0.11	0
love	0.0013	0.03	0
hate	0.0014	0.02	0
like	0.0015	0.06	0

⇨ Cross-entropy

Cross-entropy: $\mathcal{L}_i = -\log P(Y = y_i | X = x_i)$

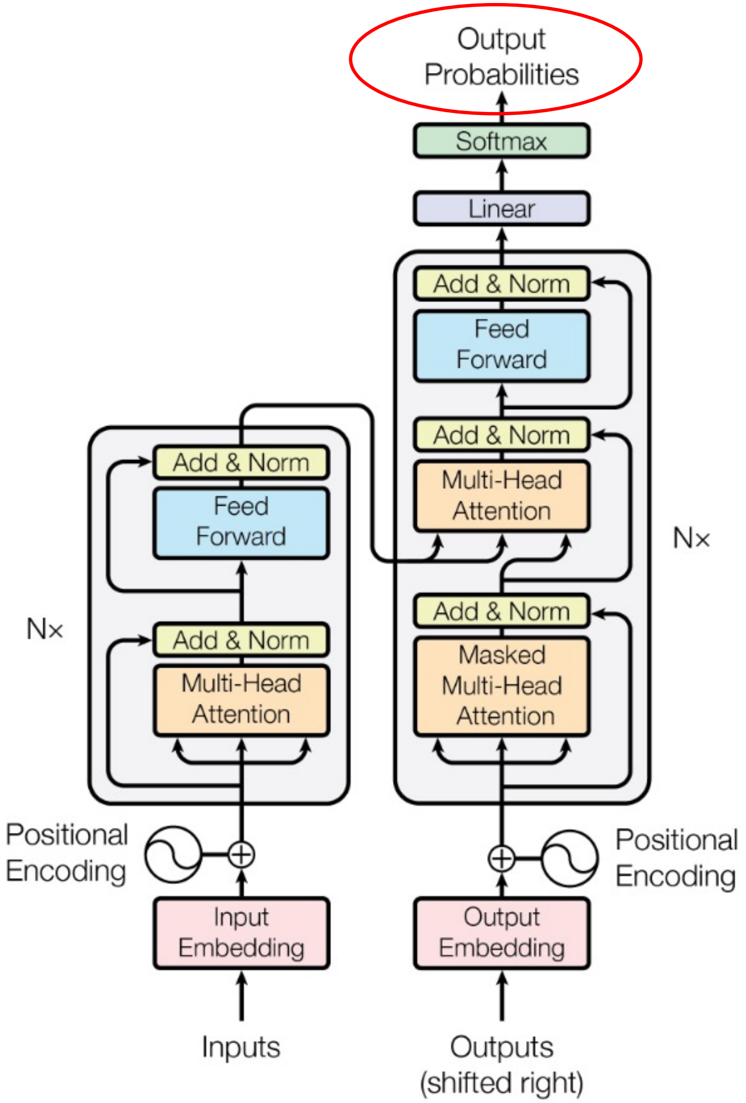
常見 Language Models

💡**Language Models: 透過 Language Modeling 去預測詞彙機率分佈的模型**

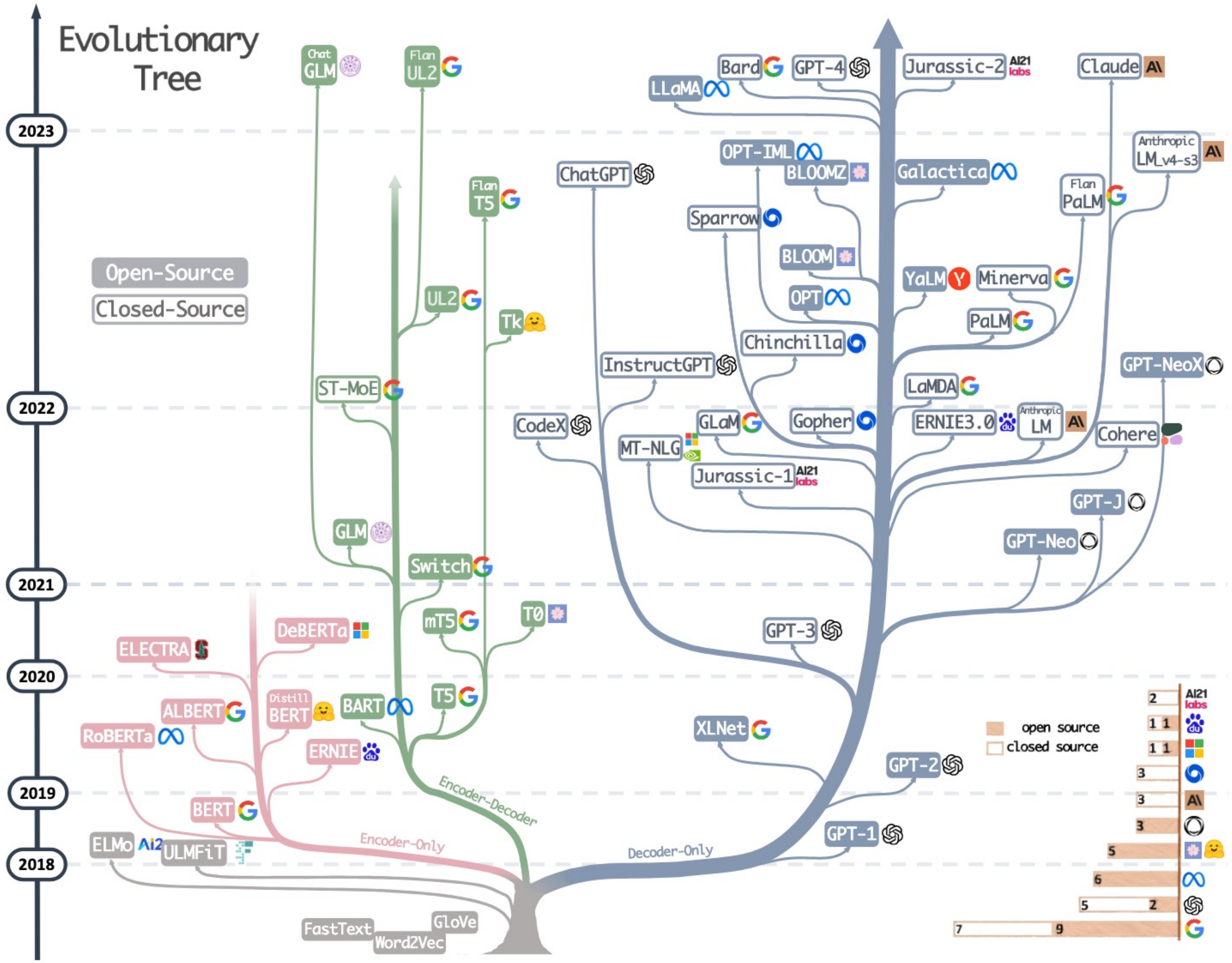
RNN

Transformer

GPT-series



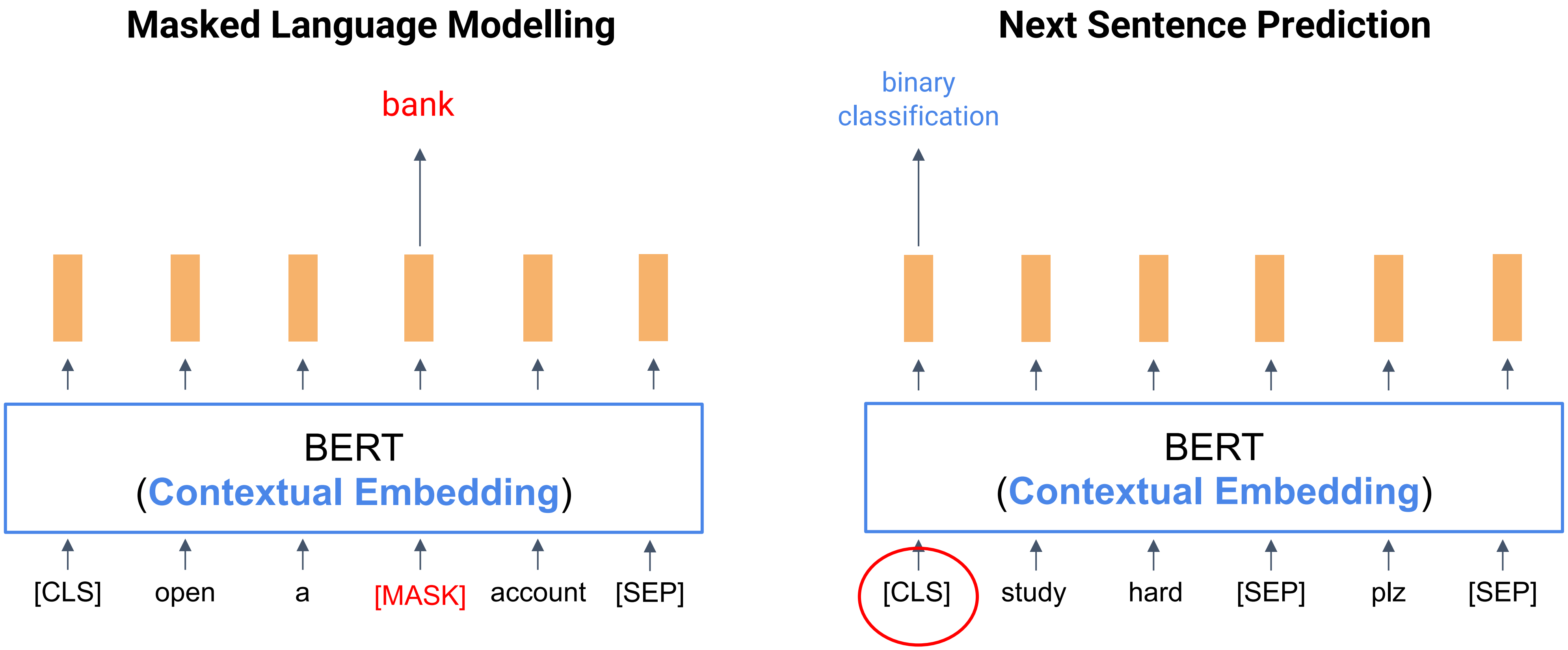
Evolutionary Tree of Large Language Models



Yang, Jingfeng, et al. "Harnessing the power of llms in practice: A survey on chatgpt and beyond." ACM Transactions on Knowledge Discovery from Data 18.6 (2024): 1-32.

BERT (Bidirectional Encoder Representations from Transformers)

- BERT 有兩種預訓練任務：



Devlin, Jacob, et al. "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding." NAACL 2019.

Next Sentence Prediction (NSP)

- 二元分類任務
- 目標：使模型能夠理解不同語句之間的關聯性

Input = `[CLS] the man went to [MASK] store [SEP]`
`he bought a gallon [MASK] milk [SEP]`

Label = `IsNext`

Input = `[CLS] the man [MASK] to the store [SEP]`
`penguin [MASK] are flight ##less birds [SEP]`

Label = `NotNext`

Summary - Key information

- Self-supervised Learning 的訓練方式是從資料中取得「答案」
  - 節省大量標註成本
  - 適用於大規模未標資料
- 影像處理領域以重建影像為 SSL 的主要方法
- 自然語言處理領域以預測文章或句子中的下一個字為 SSL 的主要方法

先 Pre-training，再 Fine-tuning

Pre-training → **Fine-tuning**

在大量資料上進行訓練，通常是自監督式 (Self-Supervised Training)

- 影像還原
- 文字還原
- 對比學習
- ...

在**目標**資料上 (Down-stream tasks, 下游任務) 進行訓練，通常是監督式 (Supervised Training)，也就是需要有標註的資料才能進行模型訓練

- 影像分類
- 文字分類
- ...

Thank you!

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